

Parameter Estimation and Optimal Control in Epidemic Modelling

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Introduction

The COVID-19 pandemic emerged in late 2019 and spread globally within months, becoming one of the most significant public health crises of the modern era.

Mathematical epidemic models provide a structured way to understand how diseases spread, estimate key transmission parameters, and evaluate the potential impact of interventions.

The SIR framework — introduced by Kermack and McKendrick (1927) — remains the foundational tool in epidemiology nearly a century later due to its tractability and interpretability.

670 M+

Confirmed Cases (Global)

6.9 M

Confirmed Deaths (Global)

~1.03%

Case Fatality Rate

3.5 M/day

Peak Daily Cases (Jan 2022)

Introduction: COVID-19 Pandemic Context

Timeline

Emerged in Wuhan, China in late 2019; declared a pandemic by WHO on March 11, 2020.

Scale

Over 670 million confirmed cases and nearly 7 million deaths globally by end of 2022.

Waves

Four distinct waves driven by Alpha, Delta, and Omicron variants with different transmissibility.

Interventions

Governments imposed lockdowns, mask mandates, and social distancing to curb spread.

Vaccines

Rollout began in December 2020 — dramatically reducing hospitalisations and deaths.

Modelling role

Mathematical models informed policy decisions on timing, intensity, and type of interventions.

Objectives

- 1 Study the spread of infectious diseases using the SIR epidemic model
- 2 Understand the roles of susceptible, infected, and recovered populations
- 3 Estimate key model parameters: transmission rate β and recovery rate γ
- 4 Compare model predictions with observed epidemic data from JHU CSSE
- 5 Analyse how changes in parameters affect the spread of disease
- 6 Study the effect of intervention strategies such as vaccination or social distancing
- 7 Compare epidemic dynamics with and without control measures
- 8 Understand how mathematical modelling supports public health decision-making

Literature Review: Compartmental Models

Foundations & Extensions

Kermack & McKendrick (1927)

Introduced the SIR framework and derived the epidemic threshold: an outbreak grows only if $R_0 > 1$. Remains the standard starting point nearly a century later.

SEIR Model Extensions

Adds an Exposed compartment for the incubation period — more appropriate for COVID-19 (median incubation ≈ 5 days). Further extensions include age structure, spatial heterogeneity, and network-based contacts.

COVID-19 R_0 Estimation

Early estimates placed R_0 between 2.2–3.6 for the original Wuhan strain. Delta ($R_0 \approx 5$ –6) and Omicron ($R_0 \approx 8$ –15) were substantially more transmissible.

Numerical Methods

Dormand & Prince (1980) introduced RK45; SciPy's odeint uses LSODA — switching between Adams and BDF methods, ideal for epidemic ODEs whose stiffness changes over time.

Optimal Control

Dating to Sethi & Staats (1978), the approach minimises a cost functional balancing infections against intervention cost via Pontryagin's Maximum Principle.

Data Collection

Johns Hopkins University CSSE Repository

The JHU CSSE COVID-19 dataset is one of the most comprehensive and widely cited open-access pandemic data sources. It was maintained from January 2020 until March 2023.

Files used:

- CONVENIENT_global_confirmed_cases.csv
- CONVENIENT_global_deaths.csv

Format: each column = country/region; each row = calendar date; values = daily new counts (not cumulative).

1,143

Days Covered

Jan

201

Countries & Territories

Mar

2020

Data Start

2023

Data End

~670 M

Total Cases

~6.9 M

Total Deaths

Data: Important Caveats

Undercounting

Confirmed case counts significantly underestimate true infections. Testing capacity varied enormously across countries and over time — especially during early waves.

Deaths Reliability

Death counts are generally more reliable than case counts but are still subject to reporting delays and definitional differences between countries.

Negative Values

The raw data occasionally contains negative daily counts due to retrospective reporting corrections. These were clipped to zero during preprocessing.

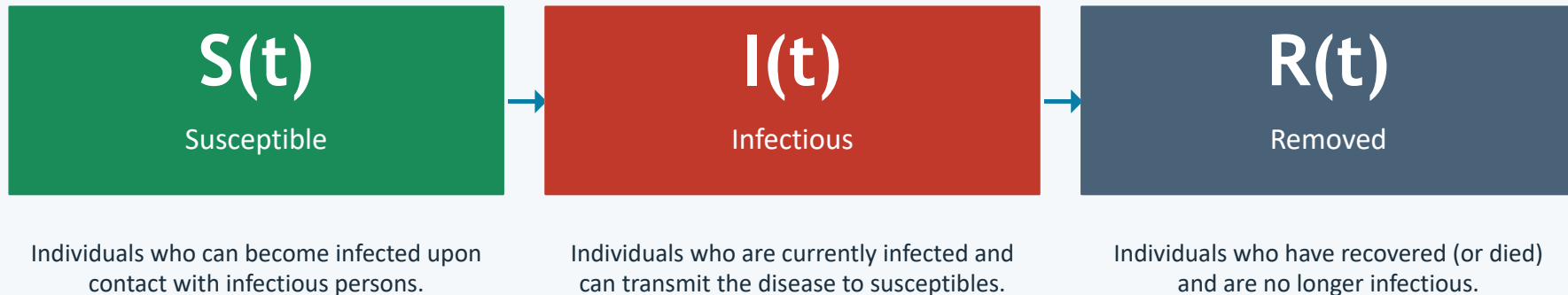
Sub-national Regions

The dataset includes sub-national regions (e.g., Australia.1 for New South Wales). These were aggregated back into national totals during preprocessing.

Rich-country Bias

High-income countries dominate the top-10 rankings partly due to superior testing infrastructure, not necessarily higher true infection burden.

Mathematical Modelling: The SIR Framework



$$\text{Conservation Law: } N = S(t) + I(t) + R(t) = \text{constant}$$

Key Assumptions:

- Homogeneous, well-mixed population — every individual can contact any other
- No births, deaths from other causes, or migration (closed population)
- Recovery confers permanent immunity — once in R, individuals do not become susceptible again

Mathematical Modelling: SIR Differential Equations

System of ODEs

$$\frac{dS}{dt} = -\beta \cdot S \cdot I$$

$$\frac{dI}{dt} = \beta \cdot S \cdot I - \gamma \cdot I$$

$$\frac{dR}{dt} = \gamma \cdot I$$

Parameter Definitions

β

Transmission rate (contacts $\times$ infection prob per day)

γ

Recovery rate ($1/\gamma$ = mean infectious period)

N

Total (constant) population size

$S \cdot I$

Rate of new infections per day

$\gamma \cdot I$

Rate of recoveries per day

Basic Reproduction Number:

$$R_0 = \beta / \gamma$$

$R_0 > 1$: spreads | $R_0 = 1$: stable | $R_0 < 1$: dies out

Mathematical Modelling: Why Numerical Methods?

The SIR system has no closed-form solution in general.

The nonlinear term βSI makes the system non-separable in time. Dividing dI/dt by dS/dt gives a relationship in the phase plane:

$$I(S) = N - S + (\gamma/\beta) \cdot \ln(S/S_0)$$

This gives the shape of the epidemic trajectory but NOT $S(t)$ explicitly. The resulting ODE for $S(t)$ contains a logarithm and is not analytically integrable — even Kermack & McKendrick could not write a closed-form $S(t)$ in their 1927 paper.

Runge-Kutta 4th Order (RK4)

$$\mathbf{y}_{n+1} = \mathbf{y}_n + (\mathbf{h}/6) (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

Global error $O(h^4)$ — highly accurate for epidemic trajectories.
Step size $h = 1$ day used throughout.

SciPy odeint / LSODA

- Automatically switches between Adams (non-stiff) and BDF (stiff) solvers
- Adapts as epidemic stiffness changes over time
- Used in the actual Python implementation

Mathematical Modelling: Parameter Estimation

Minimising Sum of Squared Residuals (Nonlinear Least Squares)

Since the SIR model predicts cumulative cases as $I(t) + R(t)$, and observed case counts y_i differ from model predictions $\hat{y}_i$ by Gaussian noise, maximising the likelihood is equivalent to minimising:

$$SSR(\beta, \gamma) = \sum_i (y_i - \hat{y}_i(\beta, \gamma))^2$$

Initial Guess:	$\beta_0 = 0.3, \gamma_0 = 0.1$ — physically plausible starting point
Bounds:	Both $\beta, \gamma \in (10^{-4}, 10)$ — prevents degenerate solutions
Algorithm:	L-BFGS-B: Limited-memory Broyden–Fletcher–Goldfarb–Shanno with box constraints
Evaluation:	At each iteration, odeint integrates the SIR system forward and compares with data
Convergence:	Fitted: $\hat{\beta} = 5.0000, \hat{\gamma} = 4.9685$, giving $R_0 \approx 1.01$

Mathematical Modelling: Epidemic Control

Modified SIR with Control Variable $u(t)$

$$dI/dt = \beta(1 - u(t)) \cdot S \cdot I - \gamma \cdot I$$

$u(t) \in [0, 1]$ — fractional reduction in contact rate due to NPIs, masking, or vaccination

$$R_{eff}(t) = \beta(1 - u) \cdot S(t) / (\gamma \cdot N)$$

$R_0 \approx 1.01$ (Global Fit)

Required control: $u > 0.01$

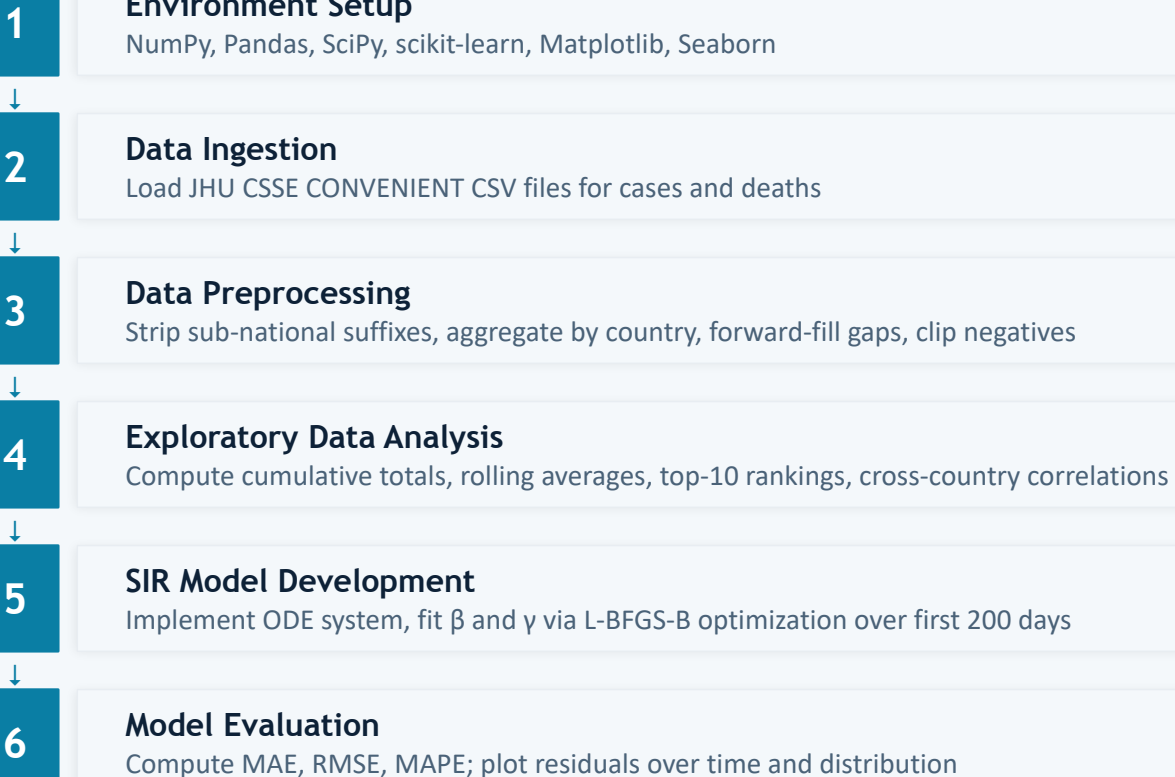
Even a 1% reduction in contacts suppresses the epidemic

$R_0 \approx 3.0$ (Country Level)

Required control: $u > 0.67$

A 67% reduction is required — consistent with strict lockdown findings

Methodology: Analysis Pipeline



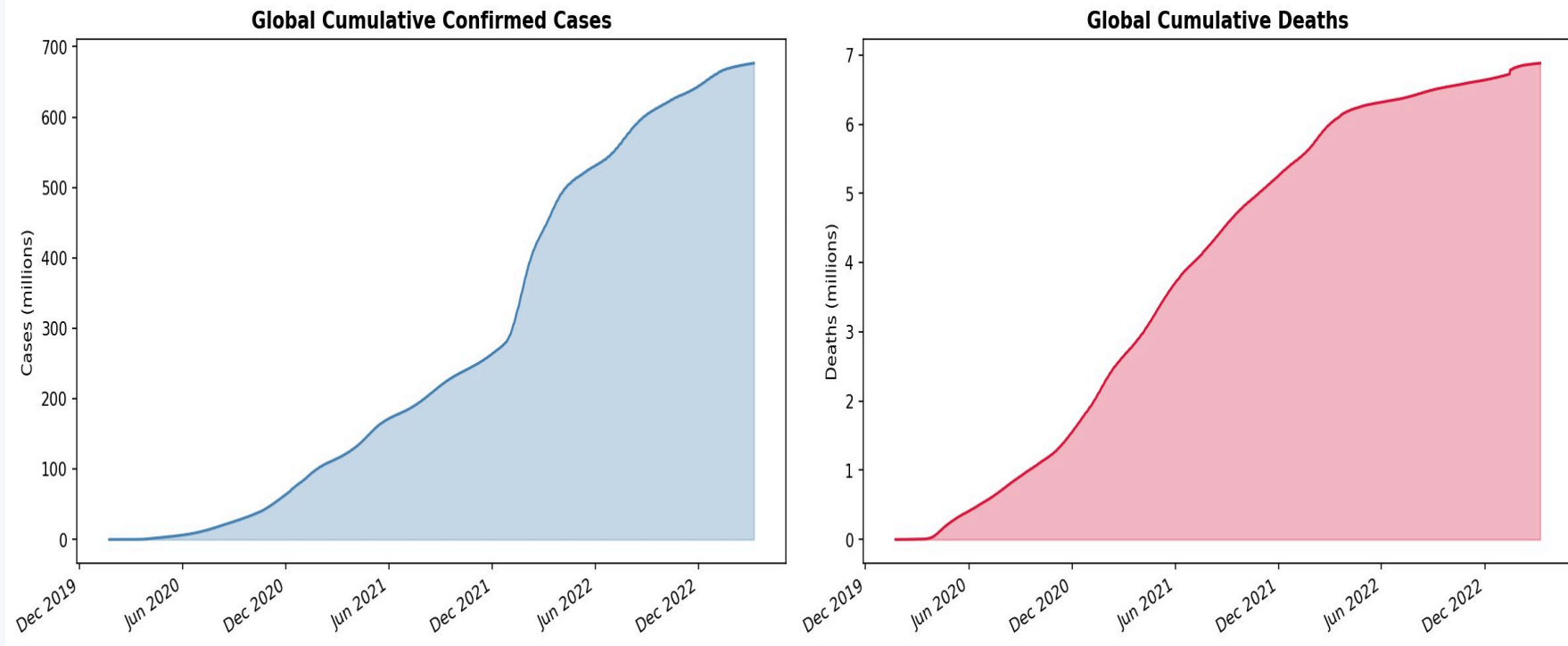
Methodology: Data Preprocessing

```
def preprocess(df):  
    dates_col = df.iloc[1:, 0].copy()  
    data_part = df.iloc[1:, 1:].copy()  
    # Strip sub-national suffixes: 'Australia.1' → 'Australia'  
    countries = [c.split('.')[0].strip() for c in df.columns[1:]]  
    data_part.columns = countries  
    data_part = data_part.apply(pd.to_numeric, errors='coerce')  
    # Aggregate sub-national regions into parent country  
    data_part = data_part.T.groupby(level=0).sum().T  
    data_part['Date'] = pd.to_datetime(dates_col.values, format='%m/%d/%y')  
    data_part = data_part.sort_values('Date').reset_index(drop=True)  
    data_part = data_part.ffill().dropna(axis=1, how='all')  
    return data_part
```

Key preprocessing steps:

Remove metadata header · Strip sub-national suffixes · Aggregate by country · Forward-fill sparse gaps · Clip negatives to zero

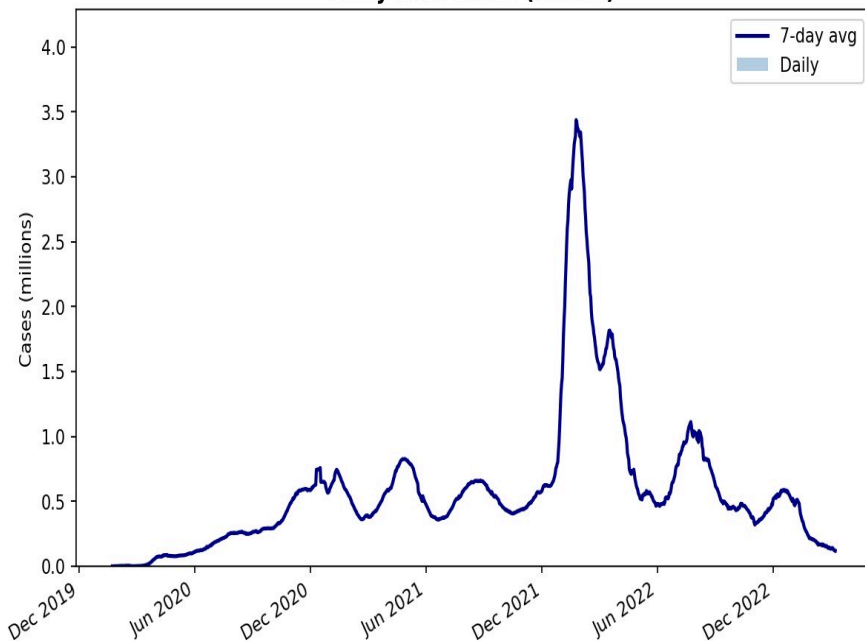
Results: Global Cumulative Trends



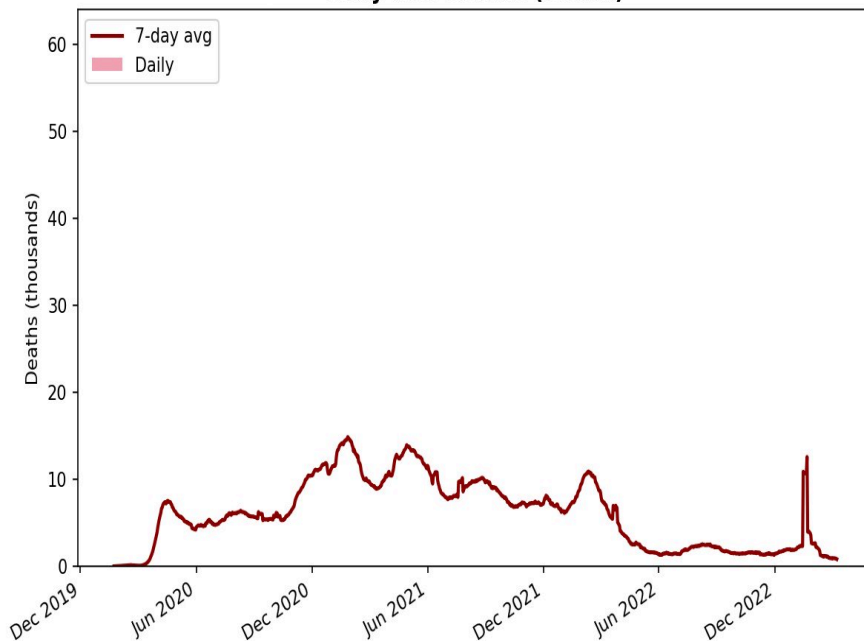
Cumulative cases reached ~670M by early 2023. Multiple growth phases correspond to successive variant-driven waves. Deaths (~6.9M) grew more gradually, reflecting improving clinical management and vaccination over time.

Results: Daily New Cases & Deaths (Global)

Daily New Cases (Global)



Daily New Deaths (Global)



Jan 2022: Omicron peak ~3.5M cases/day — largest single-day surge of the pandemic

Late 2021: Delta wave drove the highest death toll — ~15K deaths/day globally

Results: Four Epidemic Waves

Wave 1

Early 2020

Initial global spread following emergence from Wuhan. Spread driven by international travel before containment measures. First lockdowns imposed globally.

Wave 2

Late 2020

Alpha variant emerged in UK; spread across Europe and Americas during winter. Healthcare systems under severe strain. Vaccine rollout began December 2020.

Wave 3

Mid 2021

Delta variant ($R_0 \approx 5-6$) drove surges globally, including major waves in India and Southeast Asia. Higher severity than Omicron.

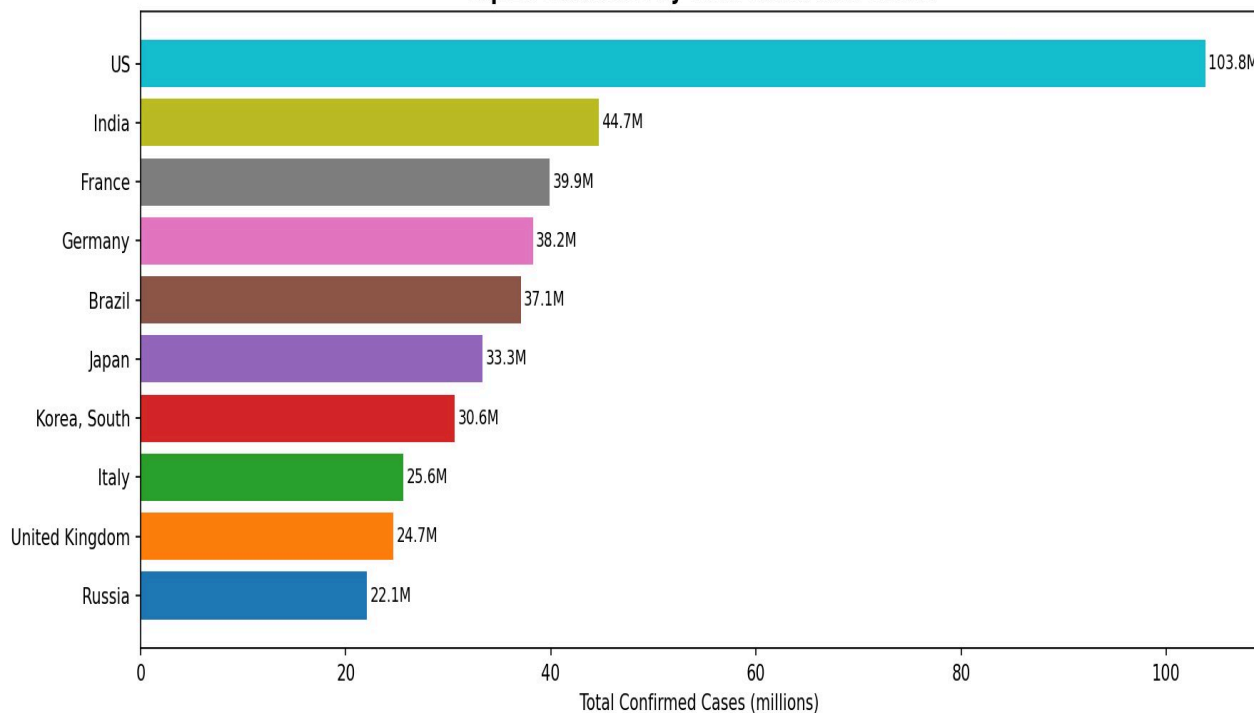
Wave 4

Jan 2022

Omicron ($R_0 \approx 8-15$) produced the largest single-day case counts ever ($\sim 3.5\text{M/day}$) but lower mortality per case due to reduced severity and high vaccination rates.

Results: Top 10 Countries by Total Confirmed Cases

Top 10 Countries by Total Confirmed Cases



US leads globally

103.8M confirmed cases — nearly 2.3× India's count

Testing bias

High-income countries dominate due to superior testing infrastructure

True burden

LMIC countries likely had higher actual infections but less diagnostic capacity

CFR variation

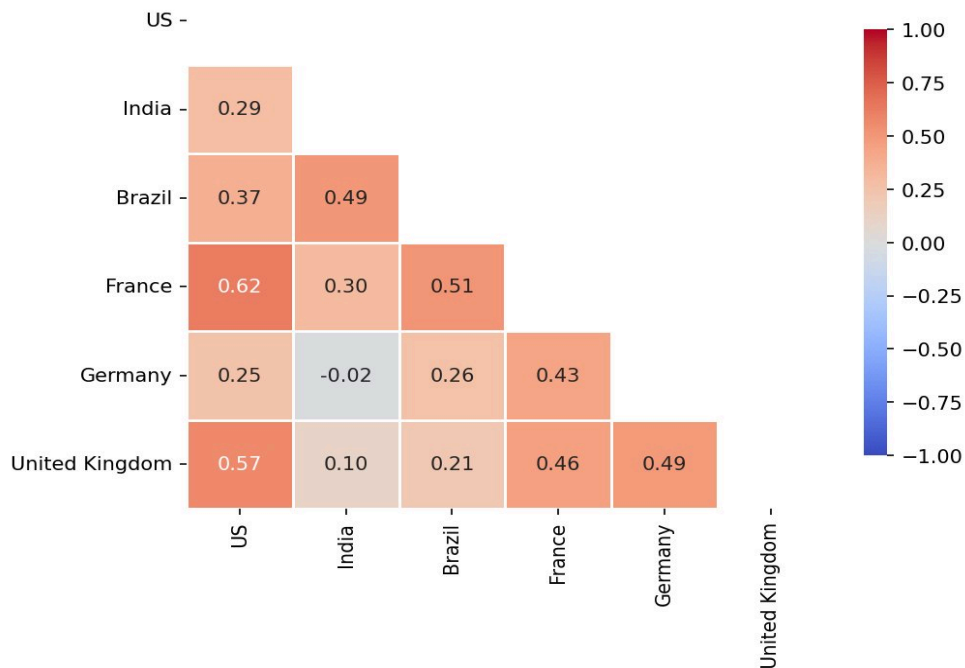
CFR ranged from <0.1% (some countries) to >3% depending on healthcare capacity

Results: Top 10 Countries – Data Summary

Country	Cases (M)	% of Global Total
United States	103.8	15.5%
India	44.7	6.7%
France	39.9	6.0%
Germany	38.2	5.7%
Brazil	37.1	5.5%
Japan	33.3	5.0%
Korea, South	30.6	4.6%
Italy	25.6	3.8%
United Kingdom	24.7	3.7%
Russia	22.1	3.3%

Results: Cross-Country Correlation of Daily Cases

Correlation of Daily New Cases
(Major Countries)



Key Findings

0.62

US ↔ France

Strongest correlation — similar epidemic timing

0.57

US ↔ UK

Shared variant introduction pathways

0.49

Brazil ↔ India

Moderate synchrony

-0.02

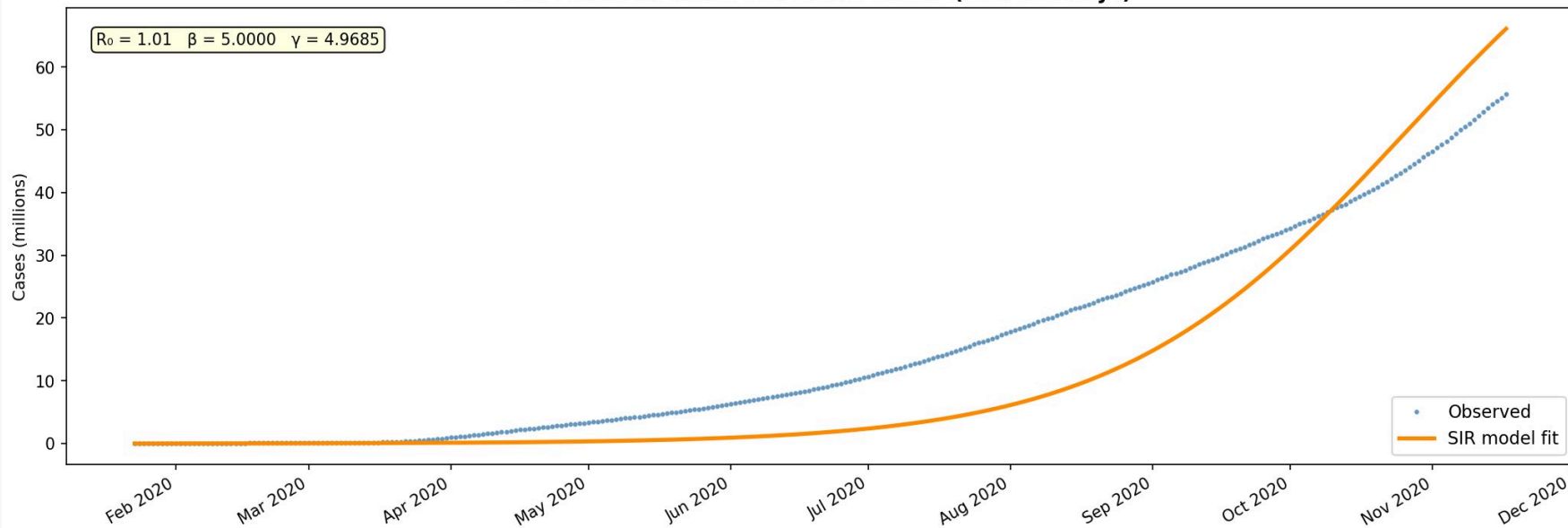
India ↔ Germany

Near-zero — distinct dynamics

Moderate correlations (0.21–0.62) suggest partial global synchrony — international travel seeds variants, but local factors

Results: SIR Model Fit (First 300 Days)

SIR Model vs. Observed Global Cases (first 300 days)

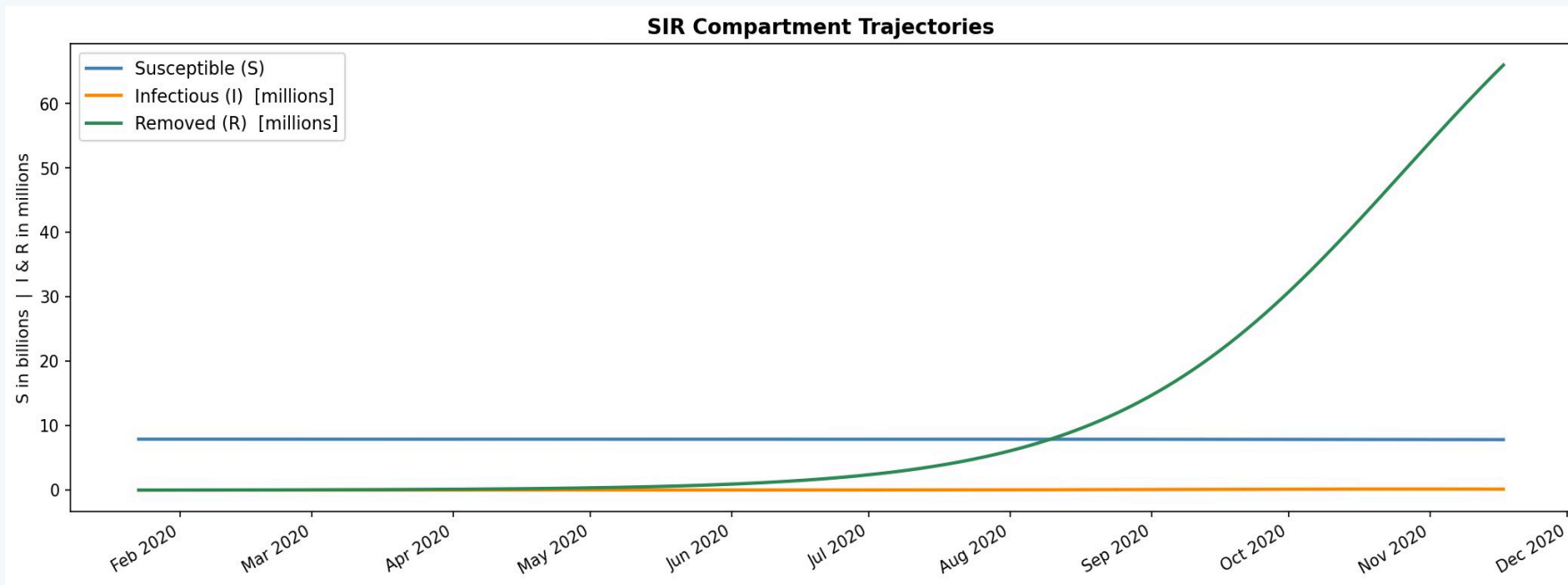


5.0000
 β

4.9685
 $\hat{\gamma}$

≈ 1.01
 R_0

Results: SIR Compartment Trajectories



Susceptible (S)

Remains ~7.7 billion throughout — only a small fraction of the global population infected by Dec 2020

Infectious (I)

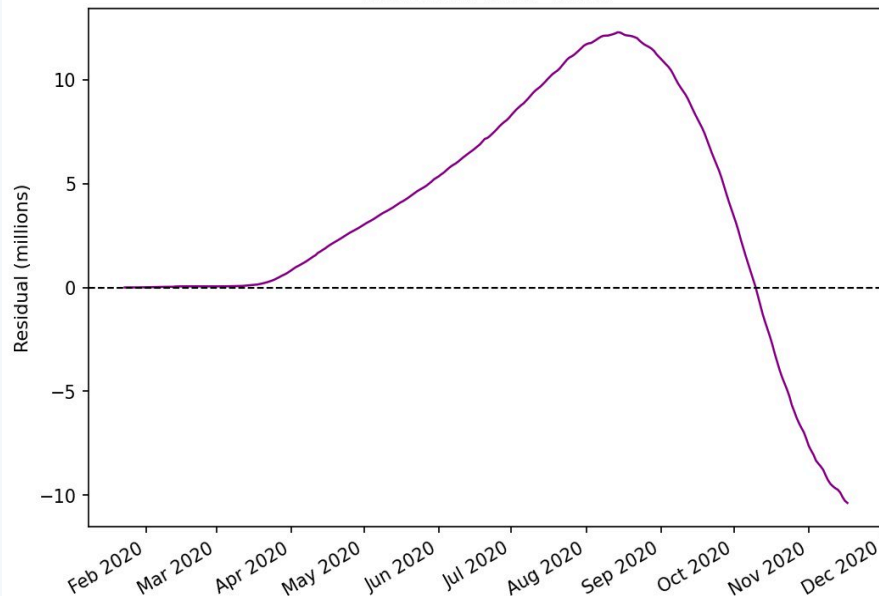
Stays small in magnitude — reflects the model's single-wave assumption and low R_0

Removed (R)

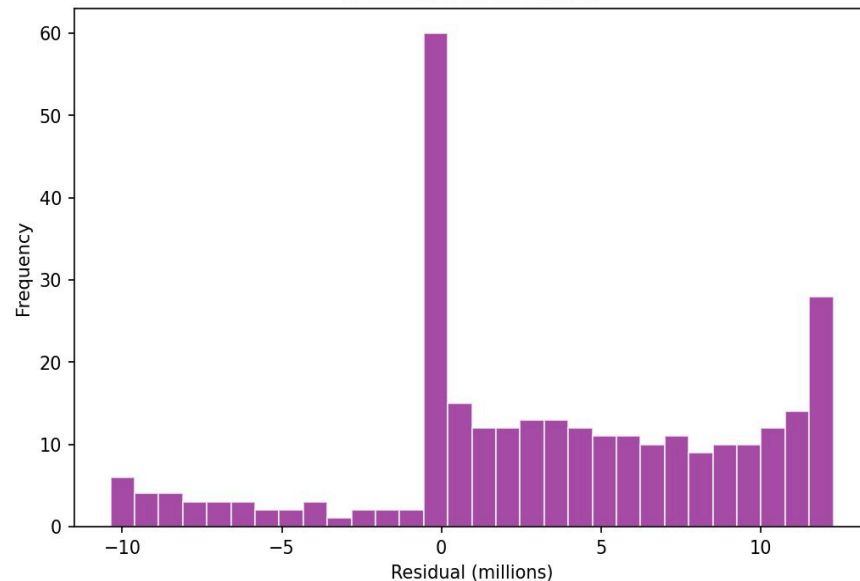
Grows steeply toward end of the 300-day window as cumulative infections accumulate

Results: Residual Analysis

Residuals Over Time



Residual Distribution



Apr-Sep 2020

Positive residuals (+12M peak): model underestimates as pandemic spreads to new regions faster than a single-wave SIR can predict

Oct-Dec 2020

Negative residuals (-10M): model overestimates as NPIs and early vaccination slow actual transmission

Results: Model Evaluation Metrics

MAE

~4.2 M cases

Mean Absolute Error — average absolute deviation between predicted and observed cumulative cases

MSE

~33.6 M²

Mean Squared Error — heavily penalises large deviations; useful for detecting outlier residuals

RMSE

~5.8 M cases

Root Mean Squared Error — same units as cases; directly interpretable

MAPE

~17.2%

Mean Absolute Percentage Error — scale-free metric; (+1 guard against zero division on early days)

Bimodal residual distribution (not centred at zero) confirms model misspecification — the basic SIR cannot capture multi-wave pandemic dynamics

Discussion: Exploratory Findings

Multi-wave Structure

Four distinct surges are visible in the daily case data: early 2020 (initial spread), late 2020 (Alpha), mid-2021 (Delta), and January 2022 (Omicron). Each wave reflects a different variant's transmissibility profile.

Omicron Paradox

Omicron produced by far the highest case counts (~3.5M/day) but comparatively fewer deaths per case — reflecting reduced intrinsic severity and widespread population immunity from prior infection and vaccination.

Rich-country Bias

High-income countries dominate the top-10 case rankings partly due to superior testing infrastructure. Low- and middle-income countries almost certainly had higher absolute infection numbers but lacked diagnostic capacity.

Partial Global Synchrony

Cross-country correlations are moderate (0.21–0.62), not high. International travel seeded variants globally, but local factors (vaccine uptake, NPIs, prior immunity, healthcare capacity) shaped each country's specific surge profile.

Discussion: Model Performance & R_0 Discrepancy

Why is the fitted $R_0 \approx 1.01$ so much lower than published estimates (2.2-3.6)?

The basic SIR assumes a homogeneous, well-mixed population with constant transmission and recovery rates. In reality, the global aggregate data blends 201 countries with heterogeneous populations, time-varying government interventions, immunity accumulation, successive variant replacements, and large cross-country testing differences. The fitted β and γ are effective averages across all these sources of variation — not true biological parameters.

⚠ Homogeneity

Assumes well-mixed population — not valid globally across 201 diverse countries

⚠ Single wave

Predicts one peak; COVID-19 had four distinct waves driven by different variants

⚠ Underreporting

Confirmed cases undercount true infections — biases parameter estimates

⚠ Constant rates

β and γ cannot vary over time — can't capture changing variants or interventions

⚠ Global averaging

Fitting global aggregate data averages out heterogeneous country-level dynamics

⚠ No immunity waning

Assumes permanent immunity — not realistic for COVID-19 over 3+ years

Discussion: Intervention Analysis

The control framework is theoretically sound despite fitting limitations.

$$\text{Reff}(t) = \beta(1 - u) \cdot S(t) / (\gamma \cdot N) \rightarrow \text{Suppression} \\ \text{when } \text{Reff}(t) < 1$$

Achievable by reducing β via NPIs (masks, distancing, ventilation) OR reducing $S(t)/N$ via vaccination.

Social Distancing

Reduces effective β by limiting contact rate. Even modest distancing ($u \approx 0.3$ – 0.5) can meaningfully delay or flatten the peak.

Mask Mandates

Reduces both the probability of infection per contact. Combined with distancing, can achieve $u \approx 0.4$ – 0.6 .

Vaccination

Reduces the susceptible fraction $S(t)/N$ directly. At sufficient coverage, achieves herd immunity even without other measures.

Strict Lockdown

Can achieve $u > 0.67$, sufficient to suppress even high- R_0 variants. Consistent with Flaxman et al. (Nature, 2020) findings.

Discussion: Natural Extensions to the SIR Model

SIR

$S \rightarrow I \rightarrow R$

Basic model: constant rates, one wave, homogeneous mixing

SEIR

$S \rightarrow E \rightarrow I \rightarrow R$

Adds Exposed compartment for incubation period
(COVID-19: ~5 days)

SEIRV

$S \rightarrow E \rightarrow I \rightarrow R$
 $S \rightarrow V$

Adds Vaccinated compartment — captures immunity from vaccination

**Time-
varying
 $\beta(t)$**

$\beta = \beta(t)$

Fits β as piecewise or spline function — absorbs changing interventions and variants

Optimal control via Pontryagin's Maximum Principle: formal derivation of $u^(t)$ that minimises a cost functional balancing total infections against economic cost of intervention*

Conclusion

Multi-wave pandemic

Global COVID-19 exhibited ~670M cases and ~6.9M deaths with four distinct waves. The Omicron surge (Jan 2022) produced the largest single-day case peak (~3.5M/day).

Cross-country correlation

Pearson correlations ranged from -0.02 (India–Germany) to 0.62 (US–France), indicating partial global synchrony with substantial local heterogeneity.

Control variable

A control $u(t)$ reduces R_{eff} proportionally. Suppression requires $u > 0.01$ for $R_0 \approx 1.01$, and $u > 0.67$ for the more realistic $R_0 \approx 3$ — consistent with empirical lockdown studies.

Country distribution

The US reported the highest total (103.8M). High-income country dominance reflects testing capacity as much as true infection burden.

SIR parameters

Fitted $\hat{\beta} = 5.0000$, $\hat{\gamma} = 4.9685$, $R_0 \approx 1.01$. Residual analysis revealed systematic non-random misfit attributable to multi-wave dynamics not capturable by a single-wave SIR.

Broader value

Despite simplifying assumptions, the SIR framework remains a tractable and interpretable foundation for public health decision-making and epidemic modelling.

Future Directions

SEIR / SEIRV Model

Add Exposed and Vaccinated compartments for greater biological realism. The SEIR model better reflects COVID-19's incubation period; SEIRV captures 2021–2022 vaccination dynamics.

Time-varying $\beta(t)$

Fit the transmission rate as a piecewise constant or smooth spline function. This allows the model to track the effects of changing interventions and variant replacements over time.

Country-Level Fitting

Fit the SIR model to individual countries rather than the global aggregate. This removes the cross-country averaging artefact and yields epidemiologically interpretable parameter estimates.

Formal Optimal Control

Apply Pontryagin's Maximum Principle to rigorously derive the optimal $u^*(t)$ that minimises a cost functional balancing total infections against the economic cost of interventions.

Bayesian MCMC

Quantify uncertainty on parameter estimates using Markov Chain Monte Carlo rather than point estimates. Provides credible intervals for β , γ , and R_0 under the posterior distribution.

Thank You!

Questions & Discussion

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